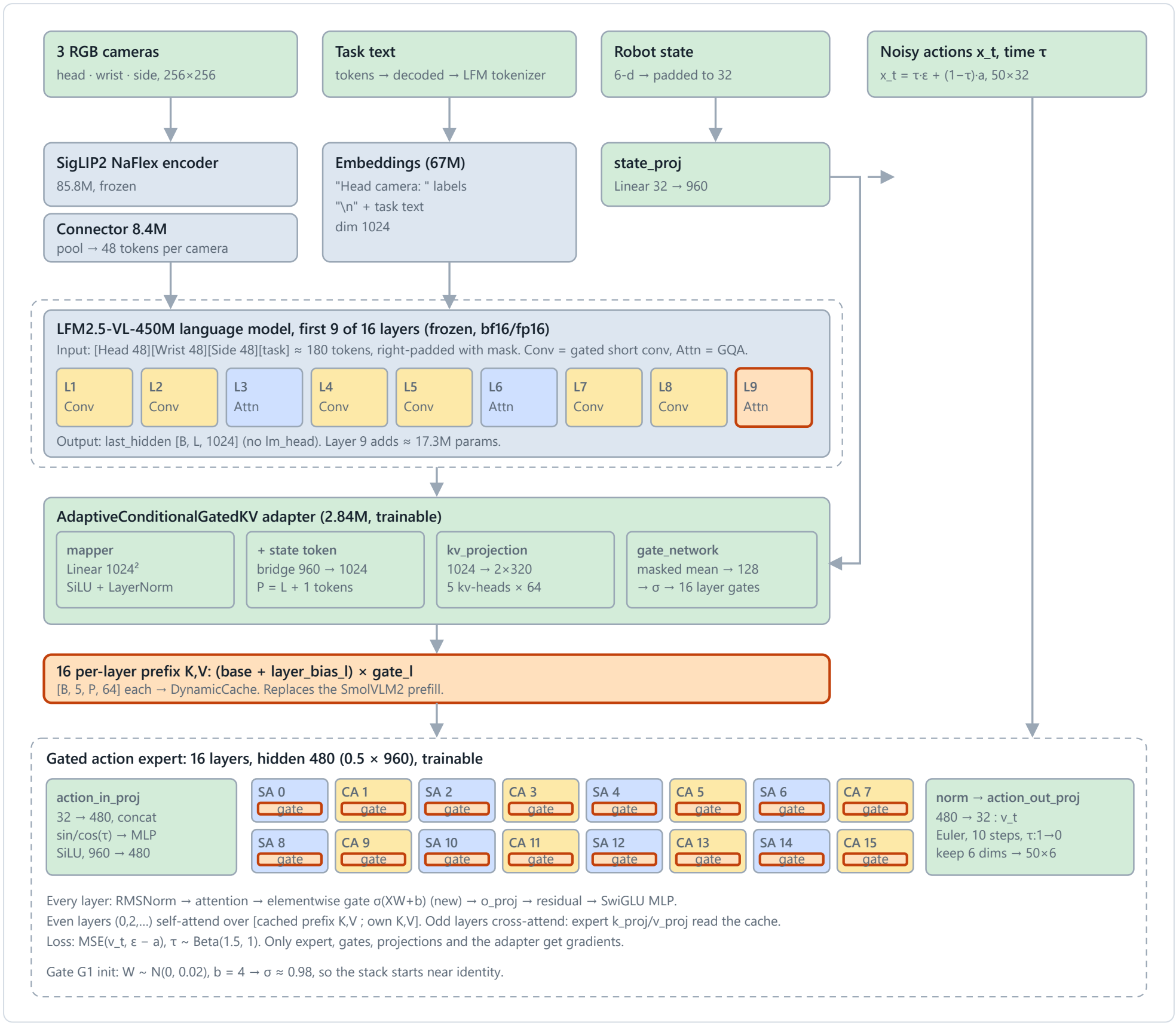
SmolVLA with an LFM2.5-VL backbone (9 layers) and gated attention

Policy from *smoller-vla.ipynb*, with one change: `LFM_KEEP_LAYERS=9` instead of 8. Everything else (adapter, gates, expert, flow-matching loss) is as in the notebook. Numbers use the training command (expert width 0.5, 3 cameras, 6-d state, chunk 50).

1. Full architecture

☐ Frozen ☐ Trainable ☐ New vs original SmolVLA / changed by the 9th layer ☐ Expert self-attention layer ☐ Expert cross-attention layer



2. Zoom: one expert layer with the gate

x (action tokens) → RMSNorm → Q from x → K, V: own (self) or from cache (cross) → SDPA, GQA 15q/5kv → $Y' = Y \odot \sigma(X_n \cdot W + b)$ → o_proj → + x → RMSNorm → SwiGLU → + residual

The gate reads the normed expert token, so it is query-dependent and can switch off individual channels per action step. It adds 16 × (480×960 + 960) ≈ 7.39M params at width 0.5.

3. Layer by layer: what is inside each block

Orange rows are new or changed relative to original SmolVLA. Parameter counts are exact: they were derived from the shapes and reproduce the 59,327,088 trainable and 365,411,376 total (8 layers) in the run log. Items marked “assumed” are not printed in the supplied sources. Kroki was not reachable (no network), so these are drawn directly in HTML.

3a. Tensor trace (batch B, 3 cameras)

3a. Tensor trace (batch B, 3 cameras)

Stage	Shape	Note
Camera frame	[B,3,256,256] in [−1,1]	→ PIL image per sample and camera → LFM image_processor
SigLIP2 + connector	[1,≈128–256,1024] per image	Token range for untiled images from LFM2 report §5.1.2
Adapter pooling	[1,48,1024] per camera	adaptive_avg_pool1d over tokens
LM input	[B,L,1024], L≈180	3×(label + 48) + task text; right-padded, mask
LM output	[B,L,1024] fp32	last_hidden_state, no lm_head
Adapter input	[B,L+1,1024]	+1 = state token
Per-layer K,V	16 × [B,5,L+1,64]	5 kv-heads × 64 = 320
Expert tokens	[B,50,480]	50-step chunk
Expert Q / attn out	[B,50,15,64] / [B,50,960]	15 q-heads, GQA 3:1
Velocity v_t	[B,50,32]	first 6 dims → actions

3b. Frozen vision path

SigLIP2 NaFlex encoder, 12 layers, d=768
85,843,200 params, frozen

patch_embedding + position_embedding
patches → 768-d; position table resized to native aspect ratio

layer_norm1
LayerNorm 768

self_attn
q/k/v_proj 768→768 with bias, bidirectional attention, out_proj 768→768

+ residual

layer_norm2
LayerNorm 768

mlp
fc1 768→3072, GELU, fc2 3072→768 (4× width assumed)

+ residual

post_layernorm (once, after layer 12)
LayerNorm 768

Connector, then token pooling
8,391,680 params, frozen (pooling has none)

PixelUnshuffle
2×2 patches merged: tokens ÷4, channels 768→3072

linear_1
3072→2048 (+bias)

activation
not shown in the sources

linear_2
2048→1024 (+bias)

adaptive_avg_pool1d (adapter)
pools the token axis to 48 per camera: a 3–5× token cut that SmolVLA does not have (it keeps 64)

3c. Frozen LFM2.5-VL language model, 9 layers

Layer	Block type	Params	Cumulative text params
Embeddings + embedding_norm	Embedding 65,536×1024		67,109,888
L1	Conv	18,355,200	85,465,088
L2	Conv	18,355,200	103,820,288
L3	Attn	17,303,680	121,123,968
L4	Conv	18,355,200	139,479,168
L5	Conv	18,355,200	157,834,368
L6	Attn	17,303,680	175,138,048
L7	Conv	18,355,200	193,493,248
L8	Conv	18,355,200	211,848,448
L9 ← added by the 9th layer	Attn (assumed)	17,303,680	229,152,128

Pattern follows the LFM2 hybrid layout: the notebook log confirms L1–L8 (conv, conv, attn, conv, conv, attn, conv, conv); L9 is assumed to be attention, check layer_types[8]. Frozen total at 9 layers: 85,843,200 + 8,391,680 + 229,152,128 = 323,387,008.

Conv layer (L1, 2, 4, 5, 7, 8)
18,355,200 params, d=1024

operator_norm
RMSNorm 1024

conv.in_proj
Linear 1024→3072, split into B, C, $\tilde{h}$

gate
$y = B \odot \tilde{h}$

Attention layer (L3, 6, and 9 if attn)
17,303,680 params, d=1024

operator_norm
RMSNorm 1024

q_proj
1024→1024 (16 heads × 64)

k_proj, v_proj
1024→512 each (8 kv-heads × 64)

conv.conv
depthwise causal Conv1d, kernel 3, 1024 channels

gate
o = C ⊙ z

conv.out_proj
Linear 1024→1024

+ residual

ffn_norm
RMSNorm 1024

feed_forward
SwiGLU: w1, w3 1024→4608, SiLU(w1 x)⊙(w3 x), w2 4608→1024

+ residual

q_layernorm, k_layernorm
QK-Norm over head_dim 64

RoPE + causal GQA attention

out_proj
1024→1024

+ residual

ffn_norm
RMSNorm 1024

feed_forward
SwiGLU: w1, w3 1024→4608, SiLU(w1 x)⊙(w3 x), w2 4608→1024

+ residual

3d. Trainable Gated-KV adapter (replaces the SmolVLM2 prefill)

AdaptiveConditionalGatedKV

2,835,216 params, trainable

mapper
Linear 1024→1024, SiLU, LayerNorm: 1,051,648

state_bridge
state_proj output 960 → Linear → 1024, appended as token P=L+1: 984,064

masked mean pool
mean over valid tokens incl. state token → [B,1024]

gate_network
Linear 1024→128, SiLU, Linear 128→16, sigmoid → g_l: 133,264

kv_projection
Linear 1024→640, chunk into base_K, base_V [B,P,320]: 656,000

layer_bias
[16,2,320] learned offsets: 10,240

for l = 0…15
K_l=(base_K+bias_l,0)·g_l, V_l=(base_V+bias_l,1)·g_l, reshape [B,5,P,64]

DynamicCache
update(K_l,V_l,l); prefix mask = [text mask, 1]

What the original did instead

SmolVLM2 prefill, 16 layers, hidden 960

Prefix tokens
[64 tok × cams][language][state token], image and text embeddings scaled by √dim

Each SmolLM2 layer
RMSNorm → GQA 15q/5kv, RoPE → RMSNorm → SwiGLU

K,V for expert layer i
taken from VLM layer i itself, after RoPE (depth-aligned)

Here
one set of final features feeds all 16 expert layers, differing only by a bias and a scalar gate; K is not rotated

3e. Trainable gated action expert, width 0.5

Self-attention layer (even 0,2…14) ×8

3,072,960 params each, d=480

input_layernorm
RMSNorm 480

q_proj
480→960 (15 heads × 64)

k_proj, v_proj
480→320 each (5 kv-heads)

RoPE
positions start at n_valid_prefix; applied to Q and own K only

KV concat
[adapter K,V layer l, P tokens ; own K,V, 50 tokens]

attention
eager GQA 3:1 in fp32, scale 1/8, mask = [prefix pad | causal over 50 actions]

gate (new)
g = σ(Linear 480→960 (normed x)); Y′ = Y ⊙ g: 461,760

o_proj
960→480; + residual

post_attention_layernorm + mlp
RMSNorm 480; gate/up 480→1280, SiLU, down 1280→480; + residual

Cross-attention layer (odd 1,3…15) ×8

2,970,560 params each, d=480

input_layernorm
RMSNorm 480

q_proj
480→960; RoPE with positions from 0

k_proj, v_proj
Linear 320→320 applied to adapter K,V of layer l (no RoPE)

attention
50 queries attend to the prefix only, mask [50×P]

gate (new)
σ(Linear 480→960) on the attention output: 461,760

o_proj
960→480; + residual

post_attention_layernorm + mlp
RMSNorm 480; gate/up 480→1280, SiLU, down 1280→480; + residual

Input and output heads

755,072 params + final norm 480

state_proj
Linear 32→960: 31,680

action_in_proj
32→480: 15,840

time embedding
sin/cos, period 0.004…4.0, dim 480; concat → 960

action_time_mlp
960→480, SiLU, 480→480: 692,160

expert norm
RMSNorm 480

action_out_proj
480→32: 15,392

Euler, 10 steps
x ← x − v/10 from τ=1 to 0; keep 6 dims

Totals (trainable)

width 0.5, 16 layers

Expert layers
8×3,072,960 + 8×2,970,560 + 480 = 48,348,640

Gates
16 × 461,760 = 7,388,160

Heads
755,072

Adapter

2,835,216

Total
59,327,088 (matches the run log)

4. Pipeline-level side by side with original SmolVLA

Original SmolVLA (smolvla_base)	This policy (LFM2.5-VL 9L + gates)
Cameras 512×512 → SigLIP + pixel shuffle 64 tokens per camera	Cameras 256×256 → SigLIP2 NaFlex + connector pooled to 48 tokens per camera
SmolVLM2-500M text model, first 16 layers hidden 960, real prefill over images + text + state token	LFM2.5-VL LM, first 9 of 16 layers (6 conv, 3 attn) hidden 1024, text only; state is not fed to it
Real K,V cache from the VLM 16 layers, one K,V per layer	Trainable adapter synthesizes the K,V cache state token, per-layer bias and gate; 2.84M params
Expert 16 layers, hidden 720 (0.75×) SA/CA alternating, no gate	Expert 16 layers, hidden 480 (0.5×), 16 elementwise gates 136 checkpoint tensors skipped, fresh init
Flow matching, 10 Euler steps → chunk of 50	Flow matching, 10 Euler steps → chunk of 50 (unchanged)

> Key conceptual change: in SmolVLA the expert reads genuine VLM keys and values. Here the LFM hidden states are re-projected by a learned adapter, so the expert reads synthetic K,V. The 9th layer only deepens the frozen feature extractor; the expert still has 16 layers.

5. Numbers

Item	Original SmolVLA	This policy (9 layers)
Backbone	SmolVLM2-500M	LFM2.5-VL-450M
Backbone layers used	16 of 32	9 of 16 (8 in the notebook)
Backbone hidden size	960	1024
Visual tokens / camera	64	48
Prefix K,V source	VLM prefill	Gated-KV adapter
Expert layers / width	16 / 720	16 / 480
Attention gates	none	16, elementwise
Total params	450.0M	≈ 382.7M (365.4M with 8 layers + 17.3M)
Trainable params	99.9M (width 0.75)	59.3M
Frozen params	350.2M	≈ 323.4M
Inference GFLOPs, batch 1	826.6 (no gates)	≈ 824 (817.7 measured at 8L, width 0.75, plus ≈ 6 for layer 9; estimate)

Layer 9 is probably a full-attention layer, going by the LFM2 hybrid layout (check `layer_types[8]`). Its parameter count is 17.3M if so, or about 18.4M if it is a conv layer. The FLOPs delta is my estimate from ≈ 180 LM tokens; the notebook only measured 8 layers.



Caveat on fairness: the trainable counts differ partly because the expert is narrower (0.5× vs 0.75×), not only because of the backbone. At width 0.5 the pretrained expert weights no longer fit, so the notebook skips them and trains those tensors from scratch.

6. Detailed comparison of the change

Basis. Original SmolVLA: full text of arXiv 2506.01844 and the lerobot source files. This policy: the notebook code and run log, plus the full LFM2 technical report for the backbone. **Question assumed** (you did not specify): what exactly changed, and what should I expect when building on it.

Row	Original SmolVLA	This policy (9 layers)
Problem addressed	Small open VLA trainable on one GPU and deployable on consumer GPUs or CPUs (abstract)	Not stated in the notebook. Apparent aim: test an LFM2.5-VL backbone and gated expert attention on the user's dataset (560 episodes, 656,950 frames)
Core idea	Frozen first-N VLM layers feed a flow-matching expert with interleaved cross/self attention (§3.1)	Same expert; VLM K,V replaced by learned K,V from LFM hidden states; sigmoid gate after attention (Qiu et al. 2025, as cited in gate_patch.py)

What is new	Layer skipping, 64 visual tokens, interleaved CA/SA, async inference, community data (§1)	LFM backbone, Gated-KV adapter, 16 gates, expert width 0.5. LFM2 itself: gated short-conv plus few GQA layers, edge-first (LFM2 §2)
Data or setup	481 community datasets, 22.9K episodes, 10.6M frames (Table 1); pretrain 200,000 steps, batch 256, 4 GPUs (§4.3)	Fine-tune from lerobot/smolvla_base, batch 64×2 GPUs=128, 4,000 steps, seed 1000, 2×T4. 344 SmolVLM2 tensors dropped and 136 expert tensors re-initialised (log)
Headline result	The authors report LIBERO avg 87.3 and Meta-World avg 57.3 for 0.45B (Table 2); SO100 avg 78.3 vs π0 61.7 and ACT 48.3 (Table 3)	None yet. Backbone only, LFM2-VL-450M vs SmolVLM2-500M (LFM2 Table 8): MMStar 40.87 vs 37.73; MMBench 55.76 vs 51.29; RealWorldQA 52.03 vs 51.37; OCRBench 657 vs 562; MME 1,230 vs 1,456; POPE 83.68 vs 85.83
Assumptions	Mid-depth frozen VLM features suffice (Table 8); community SO100 data transfers	Learned projection of frozen LFM features can stand in for real per-layer K,V; camera1–3 = head, wrist, side
Stated limits	§5.1: single-embodiment pretraining, ~23k trajectories, OCR-heavy VLM, short-horizon tasks	None in the notebook. LFM2 report §9.2: no grounding signals, no native video, images up to 512×512 directly
Code or data	Model, code and datasets released (abstract)	Local scripts; LFM2 weights open on Hugging Face (LFM2 §9.4)

6b. Component change log

Component	Original	This policy	Effect to watch
Backbone	SmolVLM2-500M, first 16 of 32 layers	LFM2.5-VL-450M, first 9 of 16 layers (56%; paper uses ½)	Different features; Table 8 shows depth helping modestly on SmolVLM2: N=8 75.0, 16 78.5, 24 79.5, 32 80.3
Image input	512×512 padded, scaled to [−1,1]	256×256 → PIL → LFM processor, per sample on CPU	Lower resolution; slower, unbatched path
Visual tokens	64 per camera	48 per camera after pooling	Fine detail may be lost
Text	Own tokens, embeddings ×√dim	Decoded and re-tokenised with the LFM tokenizer	Task text round-trips through a tokenizer
State	Token inside the VLM prefix	Bypasses the LFM; enters only via the adapter token	Table 11 (CA): prefix 80.3 vs suffix 73.3; this third variant is untested
Prefix K,V	Real, from VLM layer i, post-RoPE	Synthesised from last hidden state, no RoPE	No depth alignment; no positional signal in cached K
Expert	hidden 720, FFN 2048, no gate	hidden 480, FFN 1280, 16 gates	Table 9: ×0.5 80.3, ×0.75 77.5, ×1.00 82.3 (noisy)
Pretrained weights	All loaded	Backbone and 136 expert/projection tensors not loaded	Part of the expert restarts from scratch in 4,000 steps
Precision	bf16 loader (fp32 after gate_patch)	LFM in bf16/fp16 autocast, expert fp32	Mixed-precision features into fp32 expert
Real-time chunking	Supported through kwargs	Ignored by the replaced sample_actions	Not usable with this policy as written
Checkpoint	Includes VLM weights	Adapter only; LFM re-downloaded on load	Smaller files; depends on the hub model
Compute (bs 1)	826.61 GFLOPs inference, no gates	817.70 at 8L (width 0.75), 695.84 of it in the frozen LFM; layer 9 ≈ +6 (estimate)	About 85% of inference is the frozen backbone

6c. Comparability check

> - **Robot success rates: not directly comparable.** SmolVLA reports LIBERO, Meta-World and SO100/SO101 tasks; this policy trains on MinhDuk/Chunking_571_Skynet_combined and the log has no evaluation yet. No ranking is possible.
> - **Backbone benchmarks: comparable between the two VLMs** (same VLMEvalKit setup, LFM2 report Table 8), with mixed results. They measure complete models, not 9 frozen layers pooled to 48 tokens.
> - **Parameters and FLOPs:** the notebook profiled 8 layers at width 0.75. Width 0.5 and layer 9 figures are my derivation from the shapes; the layer-9 FLOPs are an estimate.
> - **SmolVLA ablations** (Tables 6–13) are LIBERO runs on SmolVLM2. They motivate design choices here but do not predict LFM results.

6d. Reading

What each adds. SmolVLA: a small open recipe with layer skipping, interleaved attention and async inference, with real-robot results. This policy: a different frozen backbone, an adapter that turns its features into K,V, and expert gates, with no results yet.

Agree. Frozen mid-depth features, flow-matching expert, chunk 50, 10 Euler steps, interleaved SA/CA.
Conflict. (1) SmolVLA uses true per-layer K,V; here all 16 expert layers share one feature set. (2) SmolVLA puts state in the VLM (Table 11); here it is outside the LFM. (3) Table 8 ties depth to accuracy; the 8-vs-9 layer choice has no ablation.

Cite for what. Expert, flow matching, async: SmolVLA §3.1, §3.3. Hybrid conv/GQA backbone: LFM2

§2. LFM2-VL design and token counts: LFM2 §5. VLM benchmark numbers: LFM2 Table 8. Gated attention: Qiu et al. 2025, not supplied here and unchecked.

What none answers. Whether the gates help on robot data; whether 9 LFM layers beat 16 SmolVLM2 layers for control; adapter K,V versus real K,V; the cost of unrotated cached keys; success rates on your dataset; on-device latency of this exact path (LFM2's CPU numbers are for plain LM decoding).